

Rewriting Numbers in Equivalent Forms

MA.7.NSO.1.2

I can write a rational number as a fraction, a decimal or a percentage, and use bar notation when the decimal repeats.

First lesson of the unit.

Every decimal either stops or repeats. Today we learn to tell which.

Warm-Up

On your own. Four minutes. No dividing yet.

Predict, then check

Which of these turn into decimals that never stop? Decide before you divide.

$$\frac{1}{2}$$

$$\frac{1}{3}$$

$$\frac{1}{5}$$

$$\frac{1}{6}$$

$$\frac{3}{8}$$

Write down how you decided, without dividing.

Notes

I. Every rational number's decimal stops or repeats

- A. rational number — a ratio of two integers
- B. terminating decimal — a decimal that stops
- C. repeating decimal — never stops, repeats a pattern forever
- D. **One or the other. There is no third kind.**

Notes

II. The bar covers the repeating digits only

A. **bar notation** — a bar drawn over the digits that repeat

B. **The bar covers the repeating digits and nothing else.**

C. Write the bar, not the ellipsis. Never both.

$$\frac{1}{3} = 0.333... = 0.\overline{3}$$

$$\frac{6}{11} = 0.545454... = 0.\overline{54}$$

$$\frac{13}{6} = 2.1666... = 2.1\overline{6}$$

Notes

III. Simplify first — the denominator lies until you do

- A. **6/15 looks like a fifteenth.** It is $2/5 = 0.4$, and it stops.
- B. A fraction's denominator tells you nothing at all until the fraction is in lowest terms.
- C. **Simplify, then look. Every time.**

Notes

IV. Divide, and watch the remainders

- A. **If a remainder you have already had comes back, the digits cycle forever.**
- B. **If the division reaches a remainder of 0, the decimal stops.**
- C. Denominators that gave a repeat today: 3, 6, 9, 11, 15 — and that same 15 also gave one that stopped.

Example 1

Whiteboards. Decimal first, then the second column.

Fraction	Decimal	Stops or repeats?
$\frac{1}{3}$		
$\frac{3}{8}$		
$\frac{13}{6}$		
$\frac{38}{9}$		
$\frac{17}{5}$		

Example 1

The decimals.

Fraction	Decimal	Stops or repeats?
$\frac{1}{3}$	0.333...	repeats
$\frac{3}{8}$	0.375	stops
$\frac{13}{6}$	2.1666...	repeats
$\frac{38}{9}$	4.222...	repeats
$\frac{17}{5}$	3.4	stops

Example 1

Now in bar notation.

$$\frac{1}{3} = 0.\overline{3}$$

$$\frac{13}{6} = 2.1\overline{6}$$

Example 2

Fill the left column. Do not touch the right one yet.

Fraction	Decimal
$\frac{1}{9}$	
$\frac{2}{9}$	
$\frac{3}{9}$	
$\frac{4}{9}$	
$\frac{5}{9}$	
$\frac{6}{9}$	

Predict	Decimal
$\frac{7}{9}$	
$3\frac{2}{9}$	
$17\frac{8}{9}$	

Example 2

The pattern.

Fraction	Decimal
$\frac{1}{9}$	0.111...
$\frac{2}{9}$	0.222...
$\frac{3}{9}$	0.333...
$\frac{4}{9}$	0.444...
$\frac{5}{9}$	0.555...
$\frac{6}{9}$	0.666...

The numerator IS the repeating digit.

Example 2

Now predict the right column.

Predict	Decimal
$\frac{7}{9}$	0.777...
$3\frac{2}{9}$	3.222...
$17\frac{8}{9}$	17.888...

Example 3

Divide 5 by 9 on the TI-30XA.

0.555555556

Example 2 said $5/9 = 0.555\dots$ Is the calculator wrong?

Example 3

Neither is wrong.

- A. **The screen holds ten digits. The last one is rounded.**
- B. The 6 on the end is the calculator, not the pattern.
- C. **Read the pattern from the middle of the display. Never from the last digit.**

Example 3

And Desmos disagrees with it.

Desmos shows more digits and rounds in a different place.

Whiteboards • Question 1 of 9

Boards up on three.

$$\frac{3}{8}$$

Write it as a decimal.

One value. Bar notation where it repeats.

Whiteboards · Question 1 of 9

Answer.

$$\frac{3}{8}$$

$3 \div 8 = 0.375$, and it stops

Answer: 0.375

Whiteboards • Question 2 of 9

Boards up on three.

$$\frac{3}{8}$$

Now write it as a percent.

One value. Bar notation where it repeats.

Whiteboards • Question 2 of 9

Answer.

$$\frac{3}{8}$$

$$0.375 \times 100$$

Answer: 37.5%

Whiteboards • Question 3 of 9

Boards up on three.

$$\frac{3}{9}$$

Write it as a decimal.

One value. Bar notation where it repeats.

Whiteboards • Question 3 of 9

Answer.

$$\frac{3}{9}$$

$3 \div 9 = 0.333\dots$, and it repeats

required form:

$$0.\overline{3}$$

Answer: $0.333\dots$

Whiteboards • Question 4 of 9

Boards up on three.

$$\frac{3}{9}$$

Now write it as a percent.

One value. Bar notation where it repeats.

Whiteboards • Question 4 of 9

Answer.

$$\frac{3}{9}$$

$$0.333... \times 100$$

required form:

$$33.\overline{3}\%$$

Answer: 33.333...%

Whiteboards • Question 5 of 9

Boards up on three.

$$\frac{5}{9}$$

Write it as a decimal.

One value. Bar notation where it repeats.

Whiteboards • Question 5 of 9

Answer.

$$\frac{5}{9}$$

the numerator is the repeating digit

required form:

$$0.\overline{5}$$

Answer: 0.555...

Whiteboards • Question 6 of 9

Boards up on three.

0.777...

Write it as a fraction.

One value. Bar notation where it repeats.

Whiteboards • Question 6 of 9

Answer.

0.777...

which ninth repeats a 7 ?

Answer: 7/9

Whiteboards • Question 7 of 9

Boards up on three.

1.8666...

Write it in bar notation.

One value. Bar notation where it repeats.

Whiteboards • Question 7 of 9

Answer.

1.8666...

the 8 happens once; the 6 is what repeats

required form:

1.8 $\overline{6}$

Answer: 1.8 with the bar over the 6 only

Whiteboards • Question 8 of 9

Boards up on three.

$\frac{7}{20}$

$\frac{5}{12}$

$\frac{9}{8}$

$\frac{4}{15}$

Which of these give a decimal that STOPS?

Two of the four.

Whiteboards • Question 8 of 9

Answer.

$$7/20$$

$$5/12$$

$$9/8$$

$$4/15$$

20 and 8 are 2s and 5s; 12 and 15 carry a 3

Answer: $7/20$ and $9/8$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Ana says $\frac{3}{8}$ and 0.38 are the same number. Is she right? Explain how you know.

Write your answer in sentences.

This one is written work. Show the division you used.

Whiteboards • Question 9 of 9

Answer.

Ana says $\frac{3}{8}$ and 0.38 are the same number. Is she right? Explain how you know.

$3 \div 8 = 0.375$, and 0.375 is not 0.38

Answer: No — $\frac{3}{8}$ is 0.375

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Whatever you do not finish is tonight's practice.
4. The skills for today are posted in Google Classroom.